

Student Management System using Salesforce Cloud

Complete Step-by-Step Mini Project Guide with Automation

This document explains how to build a Student Management System using Salesforce Cloud without Apex programming. The project uses Custom Objects, Fields, Tabs, Lightning Apps, and Flow Automation.

Project Objective:

To design and develop a Student Management System in Salesforce Cloud for storing and managing student records. The application includes automation that automatically updates student status as Pass or Fail based on marks.

Student Object Fields

Field Name	Data Type
Student Name	Text
Roll Number	Number
Email	Email
Phone Number	Phone
Branch	Picklist
Year	Picklist
Marks	Number
Status	Picklist

Part 1 — Create Custom Object

- Login to Salesforce Developer Edition.
- Click the gear icon in the top-right corner.
- Open Setup.
- Go to Object Manager.
- Click Create → Custom Object.
- Enter Label as Student.
- Enter Plural Label as Students.
- Set Record Name as Student Name.

- Choose Data Type as Text.
 - Enable 'Launch New Custom Tab Wizard after saving this custom object'.
 - Click Save.
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Part 2 — Create Fields

- Open the Student object.
 - Click Fields & Relationships.
 - Click New to create fields.
 - Create Roll Number field using Number datatype.
 - Create Email field using Email datatype.
 - Create Phone Number field using Phone datatype.
 - Create Branch field using Picklist datatype.
 - Add values: Computer, IT, ENTC, Mechanical, Civil.
 - Create Year field using Picklist datatype.
 - Add values: FE, SE, TE, BE.
 - Create Marks field using Number datatype.
 - Create Status field using Picklist datatype.
 - Add values: Pass and Fail.
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Part 3 — Create Custom Tab

- Go to Setup.
 - Search for Tabs.
 - Open Tabs.
 - Under Custom Object Tabs click New.
 - Select Student object.
 - Choose any tab style.
 - Click Next and Save.
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Part 4 — Create Lightning App

- Go to Setup.
- Search for App Manager.
- Open App Manager.

- Click New Lightning App.
 - Enter App Name as Student Management System.
 - Proceed through branding options.
 - Add Students, Reports, and Dashboards to navigation items.
 - Assign app to System Administrator profile.
 - Save and Finish.
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Part 5 — Create Automation using Flow Builder

- Go to Setup.
- Search for Flows.
- Open Flow Builder.
- Click New Flow.
- Choose Record-Triggered Flow.
- Select Student object.
- Set Trigger as 'A record is created or updated'.
- Click Done.
- Click the plus icon below Start.
- Select Decision element.
- Set Label as Status.
- Create Outcome Label as Pass.
- Create condition: Marks greater than or equal to 40.
- Click Done.
- Under Pass path click plus icon.
- Select Update Records.
- Set Label as Set Pass Status.
- Choose 'Use the student record that triggered the flow'.
- Set Status field value as Pass.
- Click Done.
- Under Default Outcome click plus icon.
- Add another Update Records element.
- Set Label as Set Fail Status.
- Choose 'Use the student record that triggered the flow'.
- Set Status field value as Fail.
- Click Done.
- Click Save.
- Enter Flow Name as Student Status Automation.

- Click Activate.
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Part 6 — Test the Application

- Open App Launcher.
- Open Student Management System app.
- Open Students tab.
- Click New.
- Enter student details.
- Enter marks greater than or equal to 40.
- Save record and verify Status becomes Pass.
- Create another record with marks below 40.
- Verify Status becomes Fail automatically.

Automation Logic

IF Marks $\geq$ 40 → Status = Pass

ELSE → Status = Fail

Application Architecture

User → Student Management System App → Students Tab → Student Object → Student Records → Flow Automation

Conclusion

The Student Management System was successfully developed using Salesforce Cloud without Apex programming. The application manages student records efficiently and uses Flow Automation to automatically determine student status based on marks.